

Towards More Explainable Nonlinear Dimensionality Reduction: A Feature-Driven Interaction Approach

Aeri Cho, Hyeon Jeon, Kiroong Choe, Seokhyeon Park, and Jinwook Seo

Abstract—Nonlinear dimensionality reduction (NDR) techniques are widely used to visualize high-dimensional data. However, they often lack explainability, making it challenging for analysts to relate patterns in projections to original high-dimensional features. Existing interactive methods typically separate user interactions from the feature space, treating them primarily as post-hoc explanations rather than integrating them into the exploration process. This separation limits insight generation by restricting users' understanding of how features dynamically influence projections. To address this limitation, we propose a bidirectional interaction method that directly bridges the feature space and the projections. By allowing users to adjust feature weights, our approach enables intuitive exploration of how different features shape the embedding. We also define visual semantics to quantify projection changes, enabling structured pattern discovery through automated query-based interaction. To ensure responsiveness despite the computational complexity of NDR, we employ a neural network to approximate the projection process, enhancing scalability while maintaining accuracy. We evaluated our approach through quantitative analysis, assessing accuracy and scalability. A user study with a comprehensive visual interface and case studies demonstrated its effectiveness in supporting hypothesis generation and exploratory tasks with real-world data. The results confirmed that our approach supports diverse analytical scenarios and enhances users' ability to explore and interpret high-dimensional data through interactive exploration grounded in the feature space.

Index Terms—High-dimensional data, Nonlinear Dimensionality Reduction, Explainability, Interaction Techniques, Feature Space



1 INTRODUCTION

DIMENSIONALITY REDUCTION (DR) techniques are widely used to visualize high-dimensional data by generating low-dimensional representations, i.e., projections. These projections enable analysts to visualize and investigate patterns such as clusters, outliers, and trends, helping them to better understand the intrinsic characteristics of the data [1], [2], [3], [4], [5]. Nonlinear dimensionality reduction (NDR) techniques such as t-SNE [6], UMAP [7], and LLE [8] are particularly effective in this context [2] because they capture data distributions that lie on complex nonlinear manifolds [9].

While NDR techniques reveal intricate patterns, it is difficult to understand why such patterns appear in the projection [10], [11]. In response, researchers have developed a variety of explainability mechanisms to interpret how specific patterns emerge from high-dimensional features, including feature attribution [12], gradient-based analysis [13], and projection-level annotation tools [14]. Interactivity has also been employed to support interpretation through techniques such as augmented scatterplots [15], [16], axis reconstruction [17], [18], and direct manipulations of data points [19]. Han et al. [20] further extend direct manipulation to indirectly adjust feature weights, which are then visualized through saliency maps. Although the original high-dimensional features serve as the primary medium

for explanations, they are typically used only in supplementary ways in the interaction workflow, such as delivering distributions or augmenting visualization with feature values. This disconnect limits users' ability to directly examine and understand how features influence projection structure and restricts their control over interpretation.

To enhance explainability in NDR from a new perspective, we propose a bidirectional interaction method that grounds user interactions directly in the feature space. Unlike conventional mechanisms that explain why DR algorithms produce specific outputs [12], [13], our approach enables users to interactively adjust feature weights and observe how projection changes, such as shifts in cluster formation, group separation, or other local structural modifications. This interaction helps users understand how individual features shape specific patterns in the projection, offering a feature-centric path to understanding projection behavior by enabling users to iteratively refine projections through direct control of feature weights.

Our method introduces two complementary interaction mechanisms. First, users can adjust feature weights to directly explore their influence on projections, gaining an intuitive understanding of how different features contribute to the layout. Second, users can query the system to automatically optimize feature weights, uncovering patterns. To facilitate such queries, we define two visual semantics, **Clusteredness** and **Overlap**, which guide three query types focused on cluster identification—a key task in NDR analysis [4], [21]. This approach, termed Pattern Optimization Query, reduces the manual effort of exploring diverse feature-weight combinations while enabling users to reason

- Aeri Cho, Hyeon Jeon, Kiroong Choe, Seokhyeon Park, and Jinwook Seo are with Seoul National University, Seoul, Korea. Email: {archo, hj, krchoe, shpark}@hcl.snu.ac.kr, jseo@snu.ac.kr
- Jinwook Seo is the corresponding author.

directly about projection structures.

Responsive computations are essential for interactive visual analysis, making it crucial to accelerate NDR. To overcome the computational complexity of NDR, we substitute the original algorithm with a neural network model that approximates the projection process. Expanding on Appleby *et al.* [22], our model learns feature-weighted transformations, allowing it to generate projection updates in real time without the need for repeated UMAP computations. This enables smooth and responsive projection updates during user interactions. Furthermore, we developed a visual interface, *ClusterSense*, which integrates our interaction methods with essential functionalities such as group definitions and visual summaries.

We conducted a series of evaluations to assess the effectiveness of our interaction method and visual interface. First, quantitative experiments demonstrated the model's ability to generate projections efficiently and validated the scalability of the interaction method. Second, two use cases showed how query-based interaction guides feature weight adjustments, helping users discover and interpret underlying patterns and structures in the data. Finally, a user study examined how feature-based interaction enhances explainability by supporting projections interpretation, revealing important structures in the data, and facilitating exploratory analysis. Participants formulated and validated feature-relevant hypotheses while efficiently exploring multiple projections. Furthermore, results confirmed that query-based interaction enables users to identify meaningful patterns and feature relationships with significantly less manual effort. The study also indicated that the interface is intuitive and supports both guided and exploratory analysis, even for users without prior experience.

In summary, our approach bridges the gap between feature space and NDR projections, enhancing explainability through bidirectional user interactions and neural network integration. This work makes NDR projections more interpretable, responsive, and accessible, ultimately supporting more intuitive and effective high-dimensional data analysis. The rest of the paper is organized as follows: We introduce our proposed interaction method in Sect. 3, followed by a description of the visual interface in Sect. 4. The detailed technical implementation is provided in Sect. 5. Quantitative evaluation is presented in Sect. 6, with the use case discussed in Sect. 7, and the user study reported in Sect. 8.

2 RELATED WORK

We review relevant prior works that focus on DR technique from an interactive and explainable perspective, forming a foundation for our approach.

2.1 Dimensionality Reduction Techniques

DR methods are generally categorized into two types: Linear and Nonlinear [23]. Linear methods, such as PCA [24], LDA [25] offer direct and interpretable connections between features and projections, facilitating straightforward visual analytics [26], [27]. In contrast, nonlinear methods such as UMAP [7], t-SNE [6], and Isomap [28] focus on capturing

complex data relationships, demonstrating efficacy in revealing intricate data characteristics [3], [2]. They often preserve local structure used for cluster analysis [2], while approaches like MDS [29] and MVU [30] aim to preserve global relationships by maintaining pairwise distances. Given the advantages, we adopt NDR as the primary technique.

Recent works also explore integrating machine learning techniques, especially deep neural networks, to simulate NDR. These approaches offer scalable and interactive pathways [31], [22], [32], [33], leveraging the strengths of NDR while improving efficiency and opening up new possibilities for interactive analysis.

2.2 Interactive Approaches to Dimensionality Reduction

Since static visualization cannot fully convey high-dimensional data, various interactive approaches have been proposed to explore data with DR [34]. Broadly, user interaction with DR follows three main approaches:

The first approach is a parametric interaction that modifies parameters such as feature weights to generate new projections. This is simpler with linear methods that allow inverse transformation or focus on linear subspaces [23]. Jeong *et al.* [26] propose an interactive weight steering approach for PCA, controlling feature contributions manually while Elmquist *et al.* [35] introduce an interactive subspace analysis method. Although these are pretty straightforward, they can overwhelm users with numerous choices and obscure where to begin an analysis [36]. The second approach is direct manipulation, where users move points directly in the projection [19], [37]. This provides a more visually guided form of control, yet may lead to unrealistic changes that are not faithful to the original data structure [38]. Such interactions often require inverse transforms, e.g., reconstructing axes from original features [10] or extrapolating projection changes back into the high-dimensional space [39], both of which are challenging with NDR. Parametric interactions demand that users actively engage in their analysis and understand the effect of parameter changes on resulting projections. Direct manipulation, in contrast, lets users adjust the projected points directly, though it requires additional background computations to translate the users' actions accurately. Both can complement each other [40] for user-driven insights. Dowling *et al.* [41] combine these methods, allowing symmetric interaction with feature and data space.

Beyond direct user control, an alternative approach focuses on automated projection suggestion, where the system recommends projections based on predefined criteria. Johansson *et al.* [42] explore dimensions based on user-defined quality metrics, and Lehmann *et al.* [43] suggest minimal sets of projections that maximize insights. Fujiwara *et al.* [44] follow a similar goal by optimizing feature-weighted projections to reveal hidden manifold structures, enabling comparison of diverse projection outcomes through interactive visual analysis. These methods guide users toward informative perspectives while reducing manual effort. We integrate the strengths of both user-driven interaction and automation, designing an interaction method that balances flexibility and efficiency for NDR.

2.3 Explainable Nonlinear Dimensionality Reduction

A central challenge in using NDR is making the results interpretable and understandable to users. Several methods quantify how features contribute to point placement or grouping, including gradient-based sensitivity analysis [13], Shapley values for cluster formation [12]. Local surrogate models are applied to approximate projection behavior [45], [46] or to generate locally linear explanations within partitioned regions of the projection [47], [48]. Ovcharenko *et al.* [49] further introduces a regression-based approach to estimate feature contributions at global, local, and inter-group levels.

Other approaches focus on interpreting structures within projections. DimReader [18] and Axisketcher [17] reconstruct interpretable axes through permutations or iterative axis drawing, supporting reasoning about axes. User interaction has also been leveraged for explanation, including external knowledge labeling [50], predicate-based feature identification [51], and counterfactual analysis of semantic interactions [52]. Han *et al.* [20] translate user manipulations into feature weight updates, visualizing attribution changes with saliency maps and linking interaction to feature-level explanations.

Visual augmentation techniques enhance interpretability through supplementary visualizations rather than modifying projection directly. These include compact representation of dominated feature contribution [49], set-based visualization approach for binned attribute values [11] or structural summaries [15]. Some focused on projection quality that visualize projection errors [53], structural preservation [54], or overall information including accuracy and hyperparameter effect [55]. Thijssen *et al.* [14] further support interpretation by visualizing local and global feature attribution alongside inter-group comparison.

These approaches commonly leverage original features as explanatory resources, recognizing that referencing original features provides a direct and effective means of understanding high-dimensional data. Feature-based explanations are also well-established in machine learning [56], [57], yet they typically serve as post-hoc analysis tools rather than integral components of interactive exploration. In contrast, our approach integrates feature attribution into the interactive exploration of projection spaces. Rather than explaining the internal computation of the DR algorithm, we focus on helping users understand how adjusting feature weights influences the resulting projection layouts. This feature-driven interaction approach enhances explainability by embedding explanatory features directly into user interactions and enables more intuitive and exploratory analysis about high-dimensional data.

3 FEATURE-BASED INTERACTION

In this section, we propose a novel feature-based interaction method designed to enhance the explainability of NDR techniques. Specifically, we present two interaction strategies: slider-based and query-based, allowing users to adjust feature weights. These weights are applied to the distance metric used to calculate distances between data points. By observing how changes in projections reflect adjustments in feature weights, users can infer the influence of individual

features on the layout without directly explaining the NDR method itself. The details of each approach are discussed in Sect. 3.1 and Sect. 3.2, respectively. Additionally, Sect. 3.3 explains how to integrate both interaction methods for a more comprehensive analysis workflow.

3.1 Slider-based Interaction

The slider-based interaction enables users to directly control feature weights and observe their immediate impact on projections, making it easier to explore the influence of individual features. First, NDR techniques typically pose a challenge to explainability due to the loss of meaning in reduced axes. To address this, our approach allows users to observe how changes in the original feature space affect emerging patterns in the projection, facilitating the interpretation of data structures. Second, our method enhances feature-based hypothesis testing and exploratory data analysis. By setting feature weights based on domain knowledge or specific analysis goals, users can quickly test hypotheses about feature relationships and their impact on data structures, uncovering new insights. Additionally, our method allows researchers to iterate through different feature combinations, uncover hidden structures like clusters or outliers, and refine their analysis.

3.2 Query-based Interaction

We developed a query-based exploration method that examines projections to facilitate effective interaction within the feature space. While direct interaction with the feature space has its advantages, the key challenge lies in the high dimensionality, which can make the process cumbersome and overwhelming. To address this challenge, we propose the Pattern Optimization Query (POQ), a novel approach for efficiently exploring high-dimensional data. For clarity, we define a "group" as a user-defined set of points selected through interaction, whereas a "cluster" refers to an algorithmically determined grouping of data points. These terms are used consistently in the following sections.

3.2.1 Pattern Optimization Query

POQ simplifies the exploration process by automatically identifying optimal combinations of feature weights that highlight meaningful patterns within the data. By systematically exploring the feature space, POQ assists users in the early stages of analysis or when they are uncertain about which direction to take.

Unlike direct manipulation, POQ focuses the search on feasible projections by using an optimization framework that iteratively adjusts feature weights based on predefined semantic criteria. This approach prevents inefficient exploration of arbitrary or non-existent patterns, which can occur with direct manipulation of data points. As a result, users can examine interpretable and meaningful layouts, allowing them to leverage POQ to efficiently find projections that align with their interests or hypotheses without trial and error and manual adjustments.

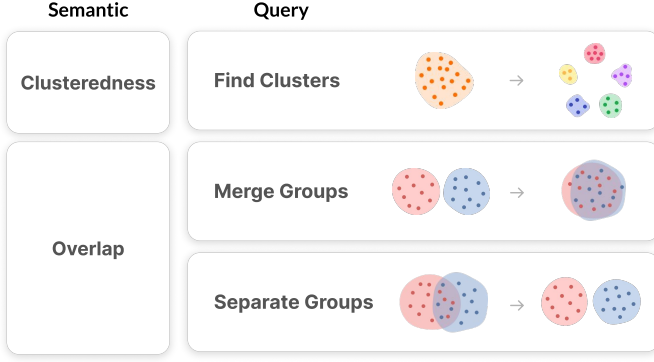


Fig. 1. Two visual semantics (left) and three types of Pattern Optimization Queries (middle) based on utilizing each semantic and the ideal result of the query in projections (right). The query searches within the feature space to generate reliable, corresponding projections and find the best fit for each semantic. (Top) **Find Clusters**: maximize **Clusteredness** seeks to present cluster structure, works with a single group. (Middle) **Merge Groups**: maximize **Overlap**, makes more points in groups become overlapped. (Bottom) **Separate Groups**: minimize **Overlap**, makes more points in groups become separated. **Merge Groups** and **Separate Groups** require at least two groups.

3.2.2 Pattern Definition

POQ leverages manifold learning’s ability to reveal data structures on a manifold, focusing specifically on local structures that NDR methods such as t-SNE and UMAP capture effectively. We define visual semantics that capture local structure across various projections by adjusting feature weights dynamically. These semantics serve as the basis for defining pattern types that guide the optimization process. This approach allows users to specify data subsets and the desired patterns, guiding them toward optimal feature weight combinations that highlight them.

We design the POQ focusing on cluster analysis, an informative feature within NDR for uncovering relationships in the high-dimensional data [4]. Our approach emphasizes cluster identification, as it allows our feature-based interaction to reveal cluster properties and relationships. As users explore the feature space and update the projection, we aim to capture key signals indicating emerging clusters. We define these signals as visual semantics, which represent noticeable changes in cluster formation. By detecting these visual semantics, users can more easily explore low-dimensional projections.

We address two analytical scenarios for identifying clusters: one where users have no specific target group and another where users focus on multiple groups. In the following section, we define the two visual semantics to assist users in identifying clusters in both scenarios and outline the corresponding queries for each.

3.2.3 Semantic: Clusteredness

The first case occurs when the cluster structures are detected within projections. When users have no specific target of interest or only a single target without a comparative group, they begin exploratory visual analysis by seeking to identify any cluster structure. This helps verify general hypotheses about the presence of clusters in the projections, providing users with a meaningful starting point to initiate further analysis.

Clusteredness is the first visual semantic we define for identifying cluster structure in this context. This fundamental structure is crucial for most analyses involving dimensionally-reduced projections [4]. Based on this, we defined **Find Clusters** (Fig. 1 Top), which maximizes **Clusteredness** to emphasize the cluster structures.

3.2.4 Semantic: Overlap

The second scenario assumes that users have groups of interest or hypotheses relevant to those groups and wish to explore relationships between them. To support this scenario, we define **Overlap** to quantify the extent of overlap among multiple groups in a projection. Specifically, overlapping groups visually suggest potential similarity or connection, whereas non-overlapping groups imply distinctiveness marking the start of cluster identification. Thus, the transition between overlapping and separate status signals a shift in the perceived relationship between the groups.

We define **Overlap** based on the principle that proximity in the DR projection suggests similarity in the high-dimensional space [38]. However, since NDR methods do not always accurately reflect distances in the high-dimensional space, interpreting “larger” distance as an absolute indicator of distinctiveness can sometimes be misleading [58], [59]. Therefore, our semantics emphasize relative changes in overlap to ensure a more reliable and meaningful distinction between groups.

Based on **Overlap**, we define two queries with opposite directions: **Merge Groups** (Fig. 1 Middle), which maximizes **Overlap**, and **Separate Groups** (Fig. 1 Bottom), which minimizes it. **Merge Groups** identifies feature weights that minimize the distinction between groups, merging them visually into a single cluster. This is useful when the user hypothesizes that the groups share similar characteristics and might be analyzed collectively. Conversely, **Separate Groups** identifies feature weights that visually separate the groups. If a user hypothesizes that the groups represent distinct clusters in the high-dimensional space with specific feature weights, **Separate Groups** can validate this by confirming their distinctiveness.

3.3 Analysis Flow with Feature-based Interaction

In this section, we summarize our proposed interaction method, highlighting how direct user adjustments and automated Pattern Optimization Query complement each other to facilitate effective exploration of the feature space.

1. Adjust Feature & Generate Group (Analysts) Initially, users can directly interact with feature space by adjusting individual feature weights to observe how each affects the resulting projections. Additionally, users can specify particular areas of interest within the dataset for targeted analysis. This process is represented by the top orange arrow, with the user in Fig. 2.

2. Select Group & Query (Analysts) Next, users can perform a Pattern Optimization Query defined in Sect. 3.2.2. These queries automatically identify cluster structures within target groups. Depending on the number of groups selected, available queries differ: **Find Clusters** is applicable to a single group or the entire dataset, while **Merge Groups** and **Separate Groups** require at least two groups.

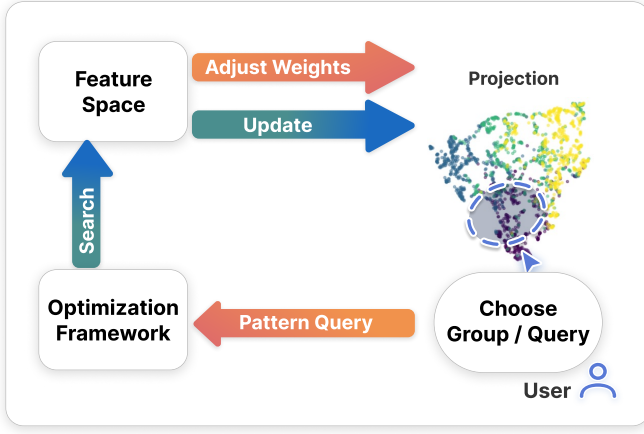


Fig. 2. Analysis flow with feature-based interaction: The projection is updated through feature weight adjustments. Users can directly adjust the weight on each feature (Top, orange) or define groups and run a Pattern Optimization Query (Bottom, orange). The optimization framework then automatically searches within the feature space and updates the projection (blue).

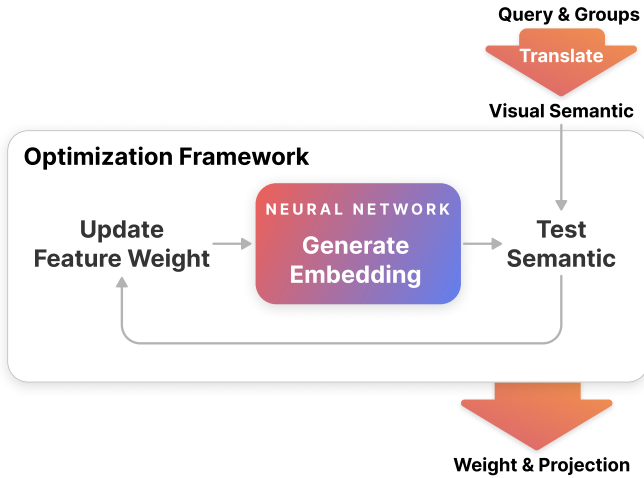


Fig. 3. Optimization Framework for automatic search : (Top) User inputs query with groups, translated into corresponding visual semantics. (Middle) The network iteratively generates projection and finds the best feature weights for the semantic. (Bottom) After optimization, the framework returns the best-fitting weights and corresponding projections.

3. Automatic Search (Optimization Framework) Once the user selects a query and target groups, the optimization framework searches for the optimal combination of feature weights that maximizes or minimizes the chosen visual semantic. This optimization process is illustrated in the middle part of Fig. 2, indicated by the blue arrows. The detailed operation of the optimization framework is depicted in Fig. 3. At the top of this figure, the user's query and selected groups are first translated into visual semantic and directional objectives. Next, in the middle of Fig. 3, the framework iteratively navigates the feature space through three repeated steps: (1) selecting candidate feature weights, (2) generating a projection using the neural network based on these weights, and (3) evaluating how well the resulting projection satisfies the specified semantic. The iterative optimization starts from the initial feature weights set by the user, refining these weights progressively

toward the optimal solution. Finally, as shown at the bottom of Fig. 3, the optimization stops after a predefined number of iterations, returning the optimal feature weights found and the corresponding projection generated by the neural network.

4. After Query (Analysts) After obtaining the query results, users can analyze the outcomes or execute another query to further explore variations in feature weights. If the resulting projection aligns with the user's intent, users can infer the characteristics of the identified patterns by examining the adjusted features. This helps users better understand cluster characteristics and the overall structures in the data, allowing users to validate their hypotheses. Since our system does not artificially enforce unattainable patterns but instead selects the most feasible feature weight combination that best approximates the requested results, users can infer when a requested pattern may not be fully achievable. In such cases, users can consider refining their hypotheses, adjusting the number of clusters or exploring alternative feature subsets to find more meaningful patterns. Regardless of the outcome, our visual interface Sect. 4 supports users by presenting visual summaries and statistical characteristics of each feature. This helps users interpret the results more effectively, validate their hypotheses, and systematically adjust their analytical strategies based on the insights gained.

In summary, our proposed interaction method guides users by providing the closest feasible results based on their queries. This approach enhances users' understanding of complex data structures by extracting meaningful information from visual patterns in low-dimensional projections. By integrating automated query results with visual interface support, our method enables users to make informed decisions and explore high-dimensional data more effectively.

4 VISUAL INTERFACE

We developed a visual interface Fig. 4 to demonstrate the applicability and effectiveness of our proposed interaction method in practical analytical scenarios. This interface integrates our novel interaction approach with essential functions, including group definition, visualization, and interaction support components. The system was developed as a web application using JavaScript, Python, and WebGL.

4.1 Prerequisites

Before using the system, it is essential to train a model, as outlined in Sect. 5 earlier. This enables exploration of the feature space and the generation of projections, either by direct feature weight adjustments or through a Pattern Optimization Query that automatically finds optimal weights matching the desired semantics.

4.2 Components

In this section, we first describe each system component and then explain how key interactions occur.

Our visual interface consists of five distinct components, organized into three functional units based on their primary roles in supporting user analysis. Each functional unit aligns

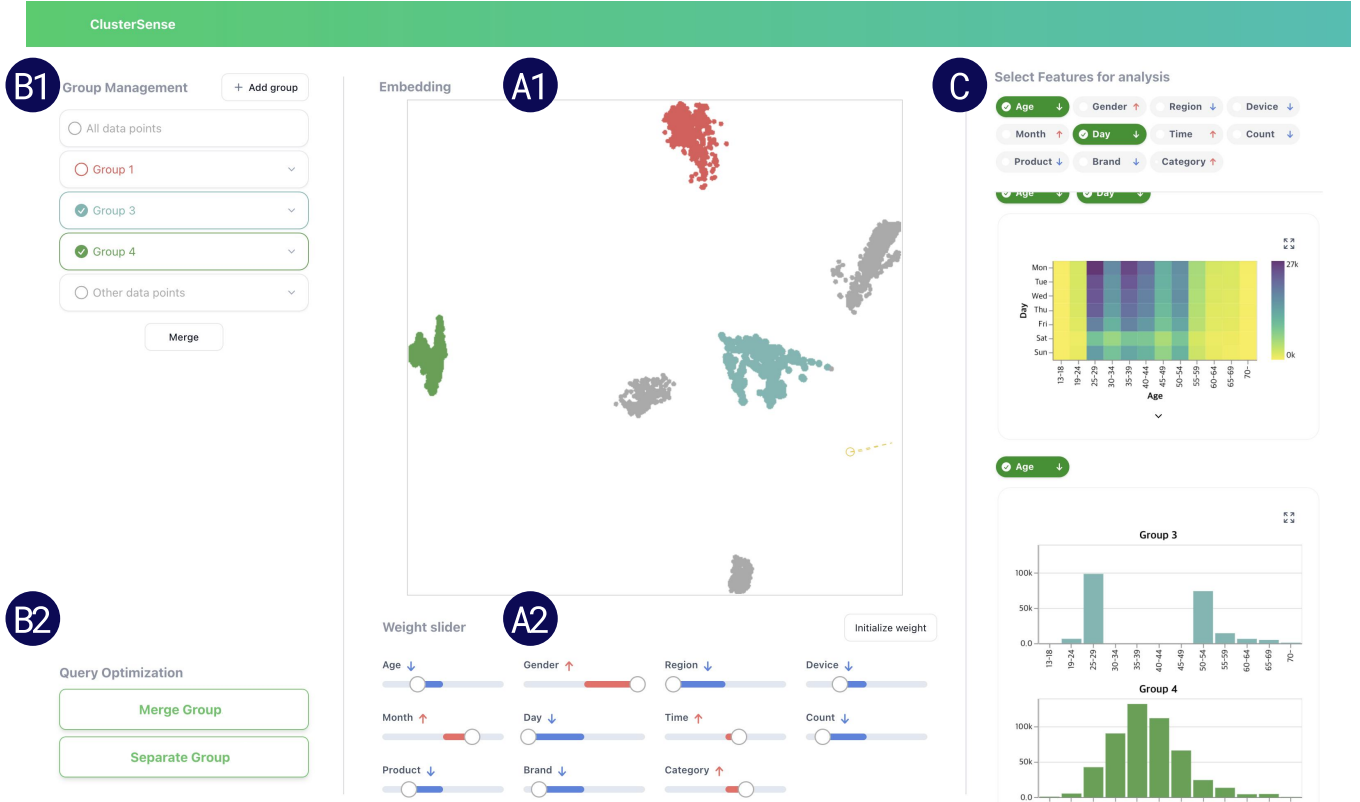


Fig. 4. **ClusterSense**: A visual interface for exploring interactive nonlinear DR empowering our pattern-based exploration strategy. (A1) Display projections corresponding to feature weights specified by the user or the query represented in (A2) (B1, B2) Manage groups of analysis targets and conduct automated query-based searches for visual analysis in NDR projections. (C) A Feature Dashboard that enables further analysis of selected groups.

with the analyst's role outlined in Sect. 3.3: (1) Adjusting feature weights, observing projections, and generating groups (2) selecting groups and performing Pattern Optimization Query based on the user's analytical goals (3) analyzing outcomes after queries, engaging in further visual analysis based on their observation and target groups. Each aligns with (1) **Adjust Feature & Generate Group**, (2) **Select Group & Query**, and (3) **After Query** in Sect. 3.3

4.2.1 Feature Weights and Embedding (A1, A2)

This unit supports **1. Adjust Feature & Generate Group**. The Embedding View Fig. 4(A1) presents the projection generated based on feature weights adjusted through sliders in the Feature Weight View Fig. 4(A2). Users can interactively adjust sliders corresponding to each feature, immediately observing changes in projections,

On the projection, users can lasso and define groups. To execute Pattern Optimization Query, the user can lock selected feature weights to their initial value. Also, after executing the Pattern Optimization Query, our optimization framework returned the desired feature weight and corresponding projection. Then, both Embedding View and Feature Weight View were updated simultaneously.

4.2.2 Specifying Groups and Query (B1, B2)

This unit supports **2. Select Group & Query**. On the left, the Group Management View Fig. 4(B1) enables users to manage target groups for analysis. In addition to lassoed

groups in the Embedding View, users can define groups by filtering data points across the entire dataset based on specific feature values. Once created, groups can be named, merged, or deleted, and they serve as the basis for Pattern Optimization Query or for visualization in the Feature Dashboard view. The Query View Fig. 4(B2), located below, allows users to execute Pattern Optimization Queries. The available query options depend on the number of groups selected: **Find Clusters** for a single group or the entire dataset, and **Merge Groups** and **Separate Groups** for two or more groups. When a query is executed, the system issues a popup warning, clarifying that the result is an approximation and represents the closest feasible pattern identified in the data.

4.2.3 Feature Dashboard (C)

This unit supports **3. After query**. Feature Dashboard Fig. 4(C) supports in-depth analysis of selected groups and features. It automatically generates visualizations based on the selected group and feature type following rules from [60] to help users explore how specific features are distributed in the groups. When users select multiple groups, the dashboard provides both summarized and individual visualizations for each group, allowing for trend identification and comparison of statistics. Visualizations can display up to two features simultaneously, visualizing their joint distributions when applicable (excluding categorical values). Additionally, users can reorder the visualizations

of features they are interested in, ensuring the most relevant information is easily accessible.

5 TECHNICAL ARCHITECTURE AND IMPLEMENTATION

In this section, we present a detailed description of the implementation behind our interaction approach.

5.1 Calculating Visual Semantic

As introduced in Sect. 3.2.2, we define two visual semantics for identifying cluster structures: **Clusteredness** and **Overlap**. These measures form the basis of our automated query interactions in later steps.

5.1.1 Clusteredness

The degree of **Clusteredness** is quantified in two steps. We first apply K-means [61] clustering algorithm to label points in the projection. The number of clusters k is chosen by the user. Second, we use the Calinski-Harabasz index [62] to quantify the resulting clustering quality, as shown in Equation 1. Here, k is the number of clusters, N is the total number of points, n_i is the number of points (size of cluster), C_i denotes the centroid of i^{th} cluster, where C is the centroid of the entire dataset.

$$Clusteredness = \frac{\sum_{i=1}^k n_i \|C_i - C\|^2}{k - 1} \times \frac{N - k}{\sum_{i=1}^k \sum_{x \in C_i} \|x - C_i\|^2} \quad (1)$$

A higher **Clusteredness** value indicates more distinct and well-separated clusters.

5.1.2 Overlap

We measure **Overlap** using *Distribution Consistency*, a distribution-based version of Class Consistency [63] that employs entropy to capture the randomness of class distributions. We assess how consistently user-defined groups appear within local regions of projection, treating these groups as cluster labels. We subdivide the projection into a 20×20 grid (400 cells) and assign each data point to a single cell. We then compute entropy in each occupied cell to evaluate how heterogeneously different groups co-occur. This grid-based scheme mitigates ambiguity of distance in NDR projections, where axes often lack inherent meaning. Instead of relying on absolute distances, *Distribution Consistency* captures local mixing, ensuring that once groups are visually distinct, further separation does not affect the score. This aligns with our goal of distinguishing merged from separated groups.

Formally, the entropy of a cell at (x, y) denoted $H(p)_{x,y}$ is given by Equation 2 and ranges from 0 to 1

$$H(p)_{x,y} = - \sum_i^m p_i \log_2 p_i, \quad p_i = \frac{n_i}{N} \quad (2)$$

where $C(X) = \{c_1, \dots, c_m\}$ represents m groups, p_i is the proportion of points n_i belonging to class c_i out of total N points in the cell.

To obtain the overall **Overlap** value, we take the sum of all cell-level entropies as shown in Equation 3.

$$Overlap = \sum_{x,y} H(p)_{x,y} \quad (3)$$

Since *Distribution Consistency* focuses on local mixing rather than absolute distances, once groups are visually distinct in the projection, further increasing their separation does not increase the score. This property fits our goal of distinguishing merged versus separated groups in the projection. A higher **Overlap** value corresponds to a lower consistency, indicating stronger local mixing among groups. This is reflected in entropy values such that a greater entropy within cells implies a more heterogeneous distribution of groups.

5.2 Neural Network for Simulating DR

To support interactive analyses, immediate feedback is crucial for understanding the connection between feature weight adjustments and the resulting projections. To achieve this, we replace UMAP with a neural network, extending Appleby *et al.* [22] This method has been shown to adjust algorithmic hyperparameters of NDR and simulate iPCA [26] by learning feature weights transformation of PCA. We extend this capability to the feature space of NDR and leverage fast inference for real-time interaction.

5.2.1 Training Data

To train the model, we generate multiple projections with different feature weights for each dataset, allowing it to learn the feature space. We randomly assign feature weights and generate between 1,000 and 5,000 projections per dataset. To isolate the effect of feature weights, we fix the DR method and its hyperparameters. We choose UMAP, a local nonlinear method known for its performance in cluster identification tasks [2]. Specifically, we keep *min_dist* and *n_neighbors* fixed to prevent additional complexity from hyperparameter variations. To determine the optimal UMAP parameters, we conduct a parameter search over *min_dist* (0.0, 0.1, 0.25, 0.5, 0.8, 0.99) and *n_neighbors* (2, 5, 10, 20, 50, 100, 200). The objective is to maximize Trustworthiness [64], which measures how well local structure in projection preserves relationships in the high-dimensional space. Detailed findings on the optimal configurations are provided in the supplementary material.

5.2.2 Model Configuration

Our model takes a concatenated vector of individual data points and feature weights as input and outputs their projected positions. This follows the HyperNP [22] framework, where the input dimension is twice the number of features, and the output is two-dimensional. Next, we explore neural network architectures, varying layer size, number of hidden layers (3 to 7), dropout rates (0.0, 0.25, or 0.5), and batch normalization. The network uses ReLU activation in intermediate layers and sigmoid function for the final layer, is trained using the Adam optimizer [65] with a Mean Squared Error loss function. The structure optimization for each dataset focused on test error using a network with maximum layer size = 2048, a total of 9 layers, batch size = 32,768, epochs = 60, and dropout rate = 0.0. Overall, the number of layers and batch normalization slightly affect performance, while lower dropout rates and a larger maximum layer size improved the result. The larger batch size is chosen because each data point is treated as an independent sample rather than the entire dataset.

5.3 Optimization Framework

We implement the Pattern Optimization Query using a neural network and optimization implementation in Optuna [66]. We employ the Tree-Structured Parzen Estimator algorithm [67], which models probability distribution to favor well-performing configurations. Also, this method is efficient and does not reduce search space due to feature correlation, ensuring flexibility. Although we use this optimization framework, other frameworks could be applied interchangeably.

6 QUANTITATIVE EXPERIMENT

In this section, we describe the quantitative experiments to evaluate the model’s performance, focusing on the accuracy and scalability of interaction. We used 15 datasets with three to 192 attributes. We used the Mac Studio (Apple M2 Max, 2023), which has a 12-core CPU, 30-core GPU, and 32GB RAM for experiments.

6.1 Model

We evaluated our model based on two key criteria: **scalability** and **simulation accuracy**. Each experiment was repeated 30 times per dataset, with random feature weights in each iteration.

Scalability Scalability is a criterion for ensuring that users receive timely feedback during interactions. Our results demonstrated a substantial speedup with model inference requiring only 16%(SD=3%) of the time required by UMAP, averaged across 15 datasets. The inference time ranged from 13.1% to 20.3% of the UMAP’s computation time across datasets, consistently accelerating projection generation. This substantial reduction in computation time validates the model’s applicability for responsive visual analytics.

Simulation Accuracy To assess how accurately network-generated projections capture the structure of high-dimensional data, we extended the parameter space of the existing [22] network architecture to include feature weights for NDR. We then evaluated the model’s ability to preserve the local structure and approximate the original UMAP output, achieving significant time savings while maintaining fidelity. This aspect is crucial because the Pattern Optimization Query relies on relationships surrounding each group or point. We followed evaluation criteria in prior works, focusing on the local structure preservation [22], [68]. Specifically, we used four key metrics: Trustworthiness and Continuity [64], which assess how well local neighborhoods are preserved during projection and vice versa, setting k as 7, following [22], [9], [69]; Mean Relative Rank Errors (MRRE) [70], which evaluates neighborhood preservation based on ranking and average scores between high-dimensional and projected spaces (average of missing and false rate) and Hit Rate [71], which measures the proportion of neighbors in the projection that share the same label. We used the Python implementation from ZADU [72]. For Trustworthiness, Continuity, and MRRE, we evaluated the simulated projection against (i) the high-dimensional data and (ii) UMAP projection, the latter to verify that our model reproduces UMAP’s local neighborhood structure. For Hit

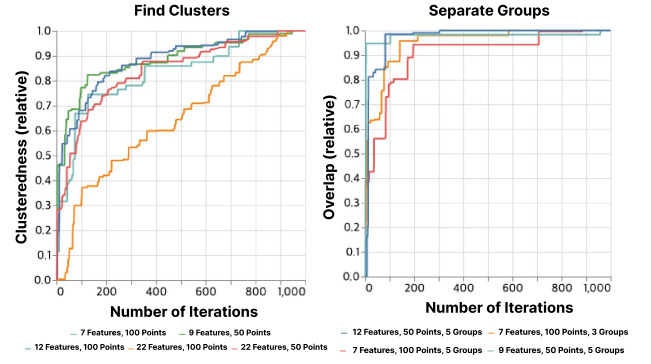


Fig. 5. **Convergence** Result for experiments under various dataset options for determining the number of iterations. The y-axis represents the score convergence, with the maximum value reached after 1000 iterations and the minimum value calculated from the initial weight.

Rate, we calculated it for both the high-dimensional data and the simulated projection and then computed Mean Absolute Error.

As a result, the average Trustworthiness, Continuity, and MRRE scores based on the high-dimensional data were 0.92 ± 0.04 , 0.94 ± 0.03 , and 0.93 ± 0.03 , respectively. When using UMAP projection as the baseline, average scores were 0.95 ± 0.04 , 0.96 ± 0.03 , and 0.94 ± 0.03 . For Hit Rate, the average MAE for each dataset was 0.027 ± 0.016 . These results demonstrate our model’s capability to simulate DR projections faithfully while preserving essential local data structures, thereby supporting insightful and meaningful data analysis.

Detailed per-dataset results, dataset characteristics, and visualizations comparing our model’s reconstructed projections with UMAP are provided in the supplementary material.

6.2 Pattern Optimization Query

We evaluated the query scalability and measured the number of iterations required for convergence.

Convergence This experiment aimed to determine the optimal number of iterations for Pattern Optimization Query to converge. Since our Optimization Framework recognizes that desired semantics may not always exist, we prioritize identifying the iteration count where semantic score stabilization occurred rather than its absolute value. We tested **Find Clusters** and **Merge Groups** using datasets with varying numbers of features and target data points. Fig. 5 shows the ratio of the semantic score at each iteration relative to the maximum semantic score achievable through 1000 iterations. After 100 iterations, the score stabilized, and the rate of changes in the score began to diminish. Since the initial semantic scores were nonzero, the absolute change was even smaller beyond this point. Based on these observations, we set the number of iterations in the Pattern Optimization Query to 100.

Scalability To assess the scalability of the query, we measured the execution time under varying dataset sizes and dimensionality. The total execution time included both projection generation and semantic evaluation. For **Find**

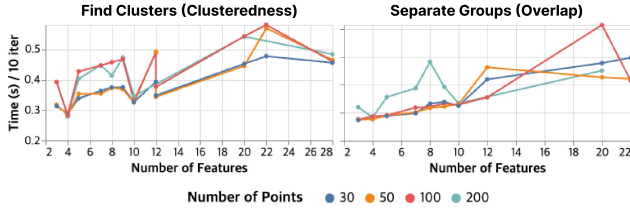


Fig. 6. Time taken per 10 iterations for **Find Clusters** (left) and **Separate Groups** (right) across varying feature sizes and target group size. The results show that the time required increases with the number of features but is unaffected by the number of target points.

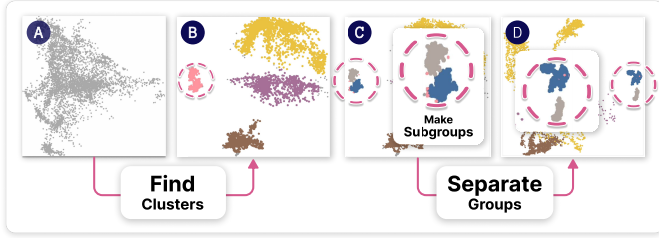


Fig. 7. **Use case 1:** (A) Initial projection and run **Find Clusters** with entire dataset. (B) Result of **Find Clusters** query. (C) Create subgroups by lassoing and run **Separate Groups** with those two subgroups (gray and blue). (D) The result of **Separate Groups** query shows that the two subgroups become slightly more distinct than before.

Clusters, we set a user input k to five, and for **Merge Groups**, we ran a query with three target groups.

As shown in Fig. 6, the results of both **Find Clusters** and **Merge Groups** indicate that while execution time increases with the number of features, it remains largely unaffected by the number of target points or groups. On average, Pattern Optimization Query with 100 iterations completed in under five seconds, which is an acceptable wait time for end users. Although the complexity of the optimization algorithm scales with the number of features, most of the query time is dominated by inference rather than the optimization process itself. Further improvements in scalability will depend on model performance, which is a crucial future avenue to explore.

7 USE CASE

We demonstrated the effectiveness and usefulness of our interaction strategy through use cases on two real-world datasets.

7.1 Case 1: Understand Customer's Behavior on Commercial Dataset

The dataset consists of online shopping purchase histories [73], containing 5,411 customer entities. Each entity has eleven categorical attributes covering purchasers' demographics (age, region, gender) and product specifics (category, quantity, brand, product). Some features are derived from continuous variables (age, time) and binary values (gender). To generate a projection, we calculated the pairwise distance for each dimension and applied feature-specific weights.

- **Iteration 1** Began with an initial projection and found no useful information (Fig. 7 A), so we performed a **Find Clusters** targeting the entire dataset, setting the cluster count k to three.
- **Iteration 2** The query updated the projection and feature weights, revealing more than three visually distinct clusters (Fig. 7 B). By defining each group and using the Feature Dashboard, we identified that Group 8 had a unique gender distribution. We then divided a group for further analysis (Fig. 7 C).
- **Iteration 3** Focusing on the subgroups from the previous iteration, we ran **Separate Groups**, which slightly increased the separation between the groups (Fig. 7 D). The query minimized overlap between groups. The Feature Dashboard revealed that categories and brands were similar across subgroups, except for one brand unique to one subgroup.

This case study demonstrates the generalizability of our system and its ability to uncover meaningful insights in complex datasets. Our findings encourage further investigation into unique purchasing behaviors and showcase the effectiveness of our system in guiding exploratory analysis and insight generation. A supplementary video illustrates the detailed interactions within our system.

7.2 Case 2: Discovering Sub Clusters and Feature Effects

The world12d [74] dataset contains 150 data points with 12 features ($F0 - F11$) across five classes (0-4). This case demonstrates how iterative feature weight interaction reveals structural patterns in the data.

- **Iteration 1** With uniform weights, most classes were well separated (Fig. 8 A), except classes 1, 2, and 3, which remained partially mixed. Applying **Separate Groups** to these classes split class 1 into a mixed and a separated part, which we defined as two groups before running **Merge Groups** (Fig. 8 B).
- **Iteration 2** Class 1 largely merged into a cluster, with notable decreases in $F8$ and $F10$ (Fig. 8 C). Testing each feature individually showed that lowering one alone did not merge the Class 1, while increasing one alone re-separated them (Fig. 8 D).
- **Iteration 3** Continuing from Fig. 8 C, we observed class 3 consistently formed two sub-clusters. We applied **Merge Groups** to merge the sub-clusters of class 3 with adjacent parts of class 1. This resulted in a large cluster containing all classes except class 0, with notable increases in $F1$ and $F2$, and decreases in $F5$ and $F7$ (Fig. 8 E).
- **Iteration 4** Within the large left cluster, class 4 appeared mostly separated from other classes during our analysis. To investigate this, we divided class 4 into a lower part (mixed with other classes) and an upper part, and then ran **Merge Groups** (Fig. 8 E). Unexpectedly, this caused class 3 to split again, with large increases in $F0$, $F7$, and $F8$. This confirmed that class 3 has a particularly strong tendency to form sub-clusters compared to other classes (Fig. 8 F).

This second use case demonstrates how our interaction strategy supports exploration of various structures within

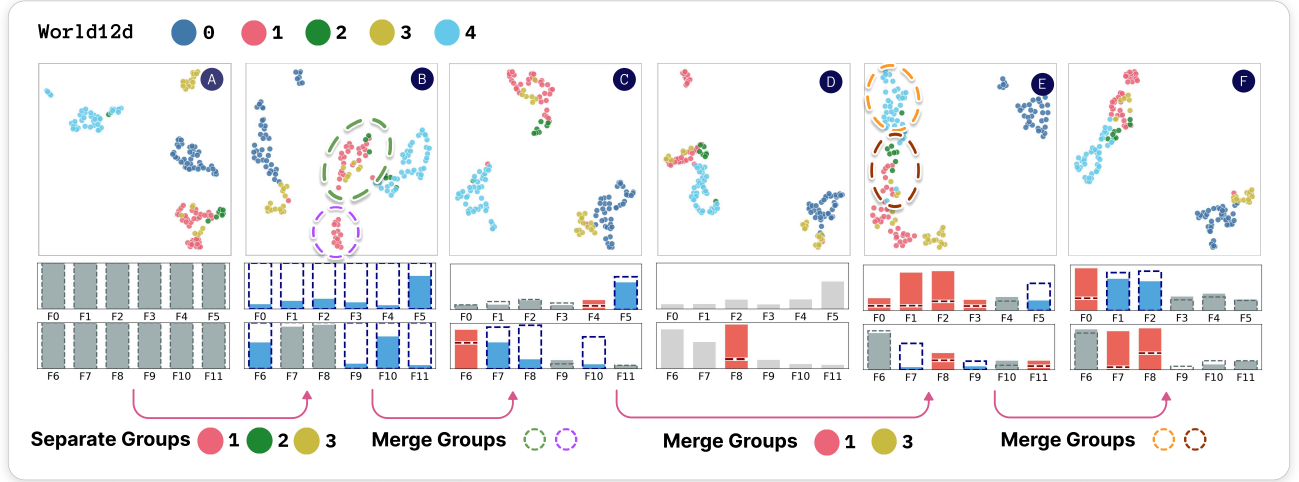


Fig. 8. **Use case 2:** (A) Initial projection with uniform weights (B) Run **Separate Groups** with classes 1, 2 and 3 (C) Run **Merge Groups** with visually separated class 1 (D) Test the individual effect of F_8 after their significant decrease (E) Run **Merge Groups** with classes 1 and 3 (F) Identify visually scattered class 4 and run **Merge Groups** on newly created groups. Below each projection, the bar chart shows the feature weights (F_0 – F_{11}), dashed boxes indicate the previous iteration's feature weight value with the vertical axis fixed to the range [0, 1]. Red bars mark increased weights, blue bars mark decreased weights while gray bars denote minor changes (difference is below 0.1)

the data. By iteratively adjusting feature weights and analyzing projection updates, users can uncover hidden sub-clusters, identify influential feature combinations, and discover class-specific behaviors not evident in the initial projection.

8 USER STUDY

We conducted a user study to evaluate the effectiveness of our interaction method (Sect. 3) and visual interface (Sect. 4) from diverse perspectives. We aimed to: (1) investigate how participants interact with our visual interface and approach analytical tasks such as pattern recognition, hypothesis generation, and validation with NDR projections. (2) understand how users utilize our interaction method and explore how bidirectional interaction with queries enhances this process within the feature space. (3) examine how individual components in the interface contribute to enriching analysis. The study received approval from our university's Institutional Review Board (IRB). All participants provided informed consent.

8.1 Experiment Setup

Participants We recruited 30 participants (8 males, 22 females) aged from 19 to 31 (24.8 ± 2.82). The participants were required to have experience in visual data analysis, specifically with scatterplots, which served as a primary visual medium for presenting DR. To ensure generalizability, participants included individuals from diverse backgrounds, including computer science, medical science, economics, and education, among others.

Data We used the same purchase history dataset as in case study 1 in Sect. 7.1, with identical projection steps. However, we adjusted the filtering criteria to increase the dataset size, allowing participants to observe a wider variety of relationships among customers during their analysis. The dataset thereby contained 7,200 points with 11 features. Since the data originated from the shopping domain,

which is closely related to daily life, we anticipated that participants could easily formulate hypotheses during the experiment.

Protocol We employed a between-subjects design to assess the impact of query-based and slider-based interaction types for feature space exploration, specifically in terms of NDR explainability. To evaluate the two interaction methods, we divided the participants into two groups: one with full interface access (QUERY) and one without the Pattern Optimization Query (NON-QUERY). Participants were labeled PQ1-PQ15 and PN1-PN15, respectively.

The study began with a 15-minute tutorial explaining the interface, interactive features, projections, and dataset. Participants then completed two analysis sessions. Each session consisted of a 10-minute analysis using the interface, where they reported observations through the think-aloud protocol, followed by a three-minute debrief to explain their overall analysis process and findings. We conducted two sessions to reduce the burden of the experiment and also to diversify our findings. After the sessions, participants completed a final 15-minute interview, providing their analysis strategies, insights, and feedback on the tool. The entire study lasted for an hour per participant. We recorded all participant interactions with the system via video and collected comments from both the debriefing and analysis sessions.

8.2 Analysis Protocol

Quantitative Analysis We investigated the effect of interaction methods on insight generation. Following Saraiya et al. [75], we defined insights as an individual observation derived from data, including not only direct answers but also exploratory questions that lead to new hypotheses.

We transcribed all participant comments and extracted statements related to the dataset, such as data patterns in a scatterplot, feature statistics, and comparisons between data subsets while excluding comments about interface

usability. Each comment was then linked to its interaction context by reviewing the video recordings to identify the corresponding interface component used.

Since our primary focus was visualization-mediated insights, we included only comments and questions derived from NDR projections. In particular, we included questions that arose from examining the projections, such as those related to pattern recognition and structure interpretation. However, we excluded questions based on participants' prior curiosity unless they explicitly referenced the visualizations. Similarly, we excluded comments generated solely using Group Management View (B1) and Feature Dashboard (C) on the interface (Fig. 4) without referencing the Embedding View. After filtering, we consolidated comments referring to the same target (e.g., a user-defined group or feature combination) into a single insight. We then classified each filtered insight into one of four categories, reflecting the analytic intent behind each comment: identifying a pattern, proposing a new explanation, verifying an assumption, or exploring feature-level details. Using transcripts from the study session, we quantified the analysis complexity by counting the number of features mentioned in each feature-based insight, excluding insights that involved only pattern recognition (later categorized as **Perception** insights in Sect. 8.3). One researcher conducted the initial labeling and counting features, followed by reviews and refinements through consensus with the other authors to minimize bias and subjectivity. To compare the QUERY and NON-QUERY conditions, we applied the Mann-Whitney U test to account for potential non-normal distributions. Effect sizes were reported using the rank-biserial correlation (r).

Qualitative Analysis We aimed to explore how participants engaged with the system and the reasons behind their interactions. We thus thoroughly examined the video recordings and transcripts from the think-aloud and debriefing sessions. To ease our analysis, we first structured the collected materials by linking comments to the relevant subparts of the video. We then grouped the insights with relevant user interactions and attempted to identify underlying reasons for participants' actions or observations based on their behavior and verbal explanations. One author conducted an initial analysis, and three more authors reviewed the results for better integrity.

8.3 Result and Discussion

Following the protocol above, we conducted quantitative and qualitative analyses to understand how participants used our feature-based interaction method. The following paragraphs discuss how their insights emerged, the complexity of their observations, and how they formed and refined hypotheses.

Our semantic coding identified four categories of insights.

- **Perception:** recognizing a specific visual pattern
 - “When I adjusted each feature, it seemed like the clusters were scattered all over.” (PQ6).
- **Hypothesis:** forming a guess about how the pattern emerged based on specific features
 - “The fact that they’re still grouped together suggests that the distribution of men might not be very diverse.” (PQ3).

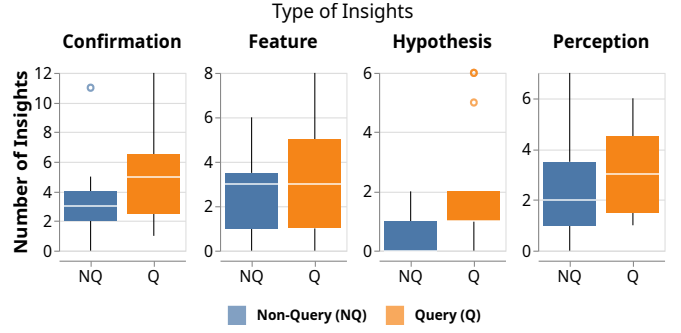


Fig. 9. Average number of insights reported under QUERY and NON-QUERY conditions by insight type. The average insight counts were similar across conditions, except for *hypotheses*, where QUERY participants reported relatively more insights.

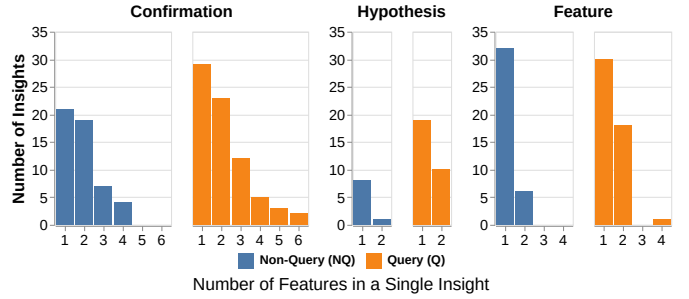


Fig. 10. Average number of features involved in a single insight under QUERY and NON-QUERY conditions, categorized by insight type (excluding *perception* insight).

- **Confirmation:** explaining the pattern by examining relevant features
 - “Although there is some overlap in regions, gender seems to be the biggest factor separating these two groups.” (PN5).
- **Feature:** identifying characteristics of a single feature or relationships between features
 - “I compared gender and region, and even a small change in gender caused a noticeable difference, so it seems gender has a stronger influence.” (PN3).

Quantitative Findings By analyzing the distribution of comments across insights types (Fig. 9), we found that the QUERY condition effectively promoted hypothesis generation. Our analysis revealed that **Hypothesis** insights were significantly more frequent under QUERY compared to NON-QUERY ($p = 0.02$, $r = 0.47$). In contrast, **Confirmation** insights did not reach statistical significance ($p = 0.081$); the effect size was $r = 0.37$.

We further examined the number of features involved per insight (Fig. 10), we found that participants in the QUERY condition utilized significantly more features in **Feature** insights, demonstrating an enhanced ability to identify relationships among multiple features ($p = 0.018$, $r = 0.23$). For **Hypothesis** insights, the difference was not statistically significant ($p = 0.190$) with the effect size ($r = 0.23$).

Qualitative Finding 1: Feature Engagement Our interaction design, grounded in the feature space, enabled participants to directly observe how their actions influenced projections. As a result, they could consistently relate pro-

jection changes to their interactions in the feature space, gaining a clear understanding of how the original high-dimensional data shaped the projections.

Participants engaged with features differently depending on the interaction method. With QUERY, participants could adjust multiple features simultaneously, leading to more diverse insights that integrated various features. PQ1 noted, *"I checked the impact of each of the most decreased features: gender, brand and weekdays (after run query), and in the end, it seems there is no correlation between brand and gender."* This ability also helped them uncover hidden relationships within the data. For example, PQ3 identified hidden similarities between the two groups by focusing on the features most influenced by the query, revealing patterns that dominant features had previously obscured, stating *"Query helped uncover information about similar groups, which would have been difficult to identify by adjusting features one by one."*

In contrast, slider-based interaction required participants to adjust features one at a time, often leading to conclusions based on a single feature. For instance, PN5 stated, *"I increased the weight of gender, but since there was no convergence, I concluded that the two groups are completely different based on gender."*

Qualitative Finding 2: Insight Generation Participants adopted different strategies for generating insights based on the interaction method. When using sliders, participants repeatedly adjusted individual feature weights to generate insights, such as identifying significant features that influenced overall data distribution. They also observed how data clusters responded to changes in feature weights. PN13 noted, *"By selecting a group and adjusting the feature weights, I could see which features influenced the formation of the cluster and how relationships between groups changed during movement."* By identifying groups of data that moved consistently, participants formed similar data groups, examined their characteristics, and compared them. Some participants specifically sought features that could effectively separate multiple groups. As PN12 explained, *"After creating groups, I examined which combination of feature weights would effectively separate these groups."*

In contrast, Query helped participants to discover serendipitous insights by enabling them to rapidly identify patterns and explore the data based on observed changes. Initially, participants used queries without specific hypotheses but then developed hypotheses based on the changes they observed. For instance, PQ8 stated *"I wanted to see how well these groups could be separated based on different attributes (with Find Clusters)."* Similarly, PQ2 mentioned, *"It would have been helpful to use the query to find clusters across the entire dataset first so that I could have started with a clearer understanding of the overall structure"*. Participants iteratively refined their insights using multiple queries in a row, leading to a complex understanding of the data. PQ7 described this approach: *"I used the query hoping it would handle the broad separation. After identifying clusters, I used subsequent queries Merge Groups and Separate Groups to know additional relationships between clusters."* As a result, it enabled a participant to uncover sub-clusters, observe group divergence, and understand which features caused these changes. Finally, the 'lock' function proved useful for find-

ing subtle patterns beyond dominant features. Participants used this function to exclude the influence of dominant features and focus on more subtle patterns in the data. PQ5 mentioned *"When running the query, I locked age, gender, and region since I had already defined these features when creating the groups. I thought it would be more useful to exclude their influence to better understand the data."*

Qualitative Finding 3: Complementary Use of Queries and Sliders Queries and sliders complemented each other in insight generation. While QUERY helped participants quickly identify patterns, sliders allowed them to verify the impact of individual features suggested by query results. For example, PQ8 noted, *"(After query) As I adjusted each slider, I wondered if this feature really had an impact."* Indeed, 11 of 15 participants with the QUERY reported that they proceeded with their subsequent exploration based on the most significantly changed features by queries. However, some participants found QUERY results difficult to interpret because multiple features were adjusted simultaneously, obscuring the impact of individual features. PQ6 mentioned, *"To see how the data distribution shifts with a single feature weight, the query makes it difficult because it adjusts multiple features at once, making it hard to understand individual characteristics."*

Interestingly, five participants in the NON-QUERY condition, despite being unaware of the query functionality, independently expressed the need for an automated approach similar to a query. For instance, PN10 mentioned, *"If there was an automated way to compare these variables, it would be helpful because manually adjusting them is overwhelming."*

Summary and Additional Findings. Combining all the findings, the results indicate that our system enables users to generate insights that are more diverse, in-depth, and unexpected. Participants adopted a strategic workflow, using queries for broad exploration, sliders to conduct detailed refinement, and the feature dashboard to validate insights in real time. The structured workflow proved accessible to participants with varying levels of visual analysis experience, ranging from those familiar with DR techniques to those with basic data analysis skills. Participants found the system intuitive and foresaw its potential applications in marketing, education, and other fields, highlighting its real-world relevance.

9 DISCUSSION

Our study demonstrates how interaction design influences users' exploration strategies and insight generation in NDR. In this section, we reflect on the contributions of our approach, discuss its limitations, and outline directions for future research.

9.1 Designing Interaction for Explainability

Our work highlights how interaction design directly shapes users' exploration strategies and insight-generation processes. By leveraging feature weights as the primary interaction medium, we guide users to validate visual findings through feature-relevant reasoning. The automated query-based interaction supports this process at scale. By allowing users to choose predefined structural goals and automatically retrieving feature weight configurations, the system

reduces the cognitive load of multi-dimensional reasoning, especially for users who are less comfortable with manual tuning.

While existing explainability methods focus on clarifying why DR algorithms produce certain outputs, our approach shows that users can develop a meaningful understanding by exploring how changes to feature weights lead to desired structures. Rather than interpreting algorithmic behavior, the system enables users to gain insights through direct manipulation and observation. This shift suggests that explainability research should treat user-driven exploration as a fundamental explanatory mechanism, not merely a supplement to algorithmic interpretation. Designing interactions for complex visualizations like NDR, however, remains challenging. The choice of interaction space—whether feature-based controls or algorithm-specific parameters—can either facilitate understanding or introduce additional complexity. Our design addresses this by centering on intuitive, feature-relevant controls that align with users’ analytical goals and narrow the Gulf of Execution [76]. This demonstrates that thoughtful interaction design is not just supportive but essential for managing complexity and improving explainability in NDR-based analysis.

9.2 Neural Networks: Benefits and Trade-offs

Incorporating the neural network is essential for achieving responsive interactions to enhance analysis effectiveness. During our user study, we observed that participants frequently refined feature weights iteratively to uncover subtle patterns. Without the speed provided by neural networks, such iterative interactions would have been impractical due to excessive waiting times. Additionally, the neural network is crucial in enabling the Pattern Optimization Query to function within a reasonable timeframe. By integrating machine learning techniques into visual analytics, our approach opens up opportunities for breakthroughs in analysis methods previously constrained by computational limitations.

However, this integration also introduces certain trade-offs. Training the model requires generating multiple projections in advance, which demands additional resources. Since our model is data-specific, any updates to the dataset necessitate re-training, adding system overhead—even if this process can be automated. Furthermore, the Pattern Optimization Query can still be another time-consuming factor in our system, which takes time in proportion to the number of target points. To further mitigate waiting times, future work could explore progressive visualization techniques that allow users to continue interacting with the system during optimization. Despite these challenges, the ability to support responsive and iterative interactions strongly justifies the use of neural networks in our approach.

9.3 Limitations and Future Works

Our interaction method has the potential to be integrated with existing explainability resources, such as augmented scatterplots [11], [51] to improve data summarization and hypothesis validation. Such integration could enhance the

applicability of our approach while providing more comprehensive and interpretable insights into high-dimensional data.

Despite its benefits, our method has limitations related to parameter choice and algorithm selection. Currently, these settings are fixed to simplify user interactions and maintain a focus on the feature-based space. However, allowing users to adjust settings such as projection methods and parameters could further enhance pattern optimization by revealing alternative structural relationships in the data.

Moreover, while we demonstrated the effectiveness of query-based interaction, there is still room for improvement. Users could benefit from guidance when they are unsure of the desired structure, for example, suggesting an appropriate number of clusters using X-means [77]. Providing multiple feature-weight combinations or clustering outcomes derived from different algorithms could help users explore a broader range of possibilities, similar to how Kwon *et al.* [78] presented and ranked clustering results from various algorithms based on quality metrics. Finally, clarifying how the query results are derived could further help users better interpret and trust the identified patterns.

Expanding query functionality beyond cluster detection—such as identifying gradual transitions across feature distributions—could also strengthen the analytical capabilities of the system. Lastly, scalability remains a key challenge. Although our query-based interaction partially addresses this issue from an interaction perspective, sliders remain essential for fine-grained exploration. Future research could explore interface-level solutions, such as grouping sliders by relevance through feature engineering, to improve usability and reduce cognitive load.

In summary, future work should aim to achieve a balanced integration of automated and manual interactions within a unified interface, offering a more powerful and seamless analytical experience.

10 CONCLUSION

We developed a novel strategy for interacting with NDR, focusing on enhancing its explainability. Our approach introduces bidirectional interaction that broadens the scope of analysis and makes the process more effective. It is grounded in feature space, which is both understandable and model-agnostic, offering clear advantages for data analysis. By incorporating a neural network for generating projections promptly and designing a visual interface, we demonstrated the practical benefits of this method, effectively supporting the whole analysis process. Our quantitative analysis and user study demonstrated that our method enables users to efficiently explore and interpret high-dimensional data, making NDR projections more intuitive and accessible. Ultimately, this work contributes toward addressing the inherent explainability challenges in NDR, supporting richer insights and more confident interpretations.

REFERENCES

- [1] L. G. Nonato and M. Aupetit, “Multidimensional projection for visual analytics: Linking techniques with distortions, tasks, and layout enrichment,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 25, no. 8, pp. 2650–2673, 2018.

- [2] J. Xia, Y. Zhang, J. Song, Y. Chen, Y. Wang, and S. Liu, "Revisiting dimensionality reduction techniques for visual cluster analysis: an empirical study," *IEEE Transactions on Visualization and Computer Graphics*, vol. 28, no. 1, pp. 529–539, 2021.
- [3] H. Jeon, H.-K. Ko, J. Jo, Y. Kim, and J. Seo, "Measuring and explaining the inter-cluster reliability of multidimensional projections," *IEEE Transactions on Visualization and Computer Graphics*, vol. 28, no. 1, pp. 551–561, 2021.
- [4] M. Brehmer, M. Sedlmair, S. Ingram, and T. Munzner, "Visualizing dimensionally-reduced data: Interviews with analysts and a characterization of task sequences," in *Proceedings of the Fifth Workshop on Beyond Time and Errors: Novel Evaluation Methods for Visualization*, 2014, pp. 1–8.
- [5] E. Kandogan, "Visualizing multi-dimensional clusters, trends, and outliers using star coordinates," in *Proceedings of the seventh ACM SIGKDD international conference on Knowledge discovery and data mining*, 2001, pp. 107–116.
- [6] L. Van der Maaten and G. Hinton, "Visualizing data using t-sne," *Journal of machine learning research*, vol. 9, no. 11, 2008.
- [7] L. McInnes, J. Healy, and J. Melville, "Umap: Uniform manifold approximation and projection for dimension reduction," *arXiv preprint arXiv:1802.03426*, 2018.
- [8] S. T. Roweis and L. K. Saul, "Nonlinear dimensionality reduction by locally linear embedding," *science*, vol. 290, no. 5500, pp. 2323–2326, 2000.
- [9] M. Espadoto, R. M. Martins, A. Kerren, N. S. Hirata, and A. C. Telea, "Toward a quantitative survey of dimension reduction techniques," *IEEE transactions on visualization and computer graphics*, vol. 27, no. 3, pp. 2153–2173, 2019.
- [10] H. Kim, J. Choo, H. Park, and A. Endert, "Interaxis: Steering scatterplot axes via observation-level interaction," *IEEE transactions on visualization and computer graphics*, vol. 22, no. 1, pp. 131–140, 2015.
- [11] J.-T. Sohns, M. Schmitt, F. Jirasek, H. Hasse, and H. Leitte, "Attribute-based explanation of non-linear embeddings of high-dimensional data," *IEEE Transactions on Visualization and Computer Graphics*, vol. 28, no. 1, pp. 540–550, 2021.
- [12] W. E. Marcilio-Jr and D. M. Eler, "Explaining dimensionality reduction results using shapley values," *Expert Systems with Applications*, vol. 178, p. 115020, 2021.
- [13] S. Corbugy, R. Marion, and B. Fréney, "Gradient-based explanation for non-linear non-parametric dimensionality reduction," *Data Mining and Knowledge Discovery*, vol. 38, no. 6, pp. 3690–3718, 2024.
- [14] J. Thijssen, Z. Tian, and A. Telea, "Interactive tools for explaining multidimensional projections for high-dimensional tabular data," *Computers & Graphics*, vol. 122, p. 103987, 2024.
- [15] K. Eckelt, A. Hinterreiter, P. Adelberger, C. Walchshofer, V. Dhanoa, C. Humer, M. Heckmann, C. Steinparz, and M. Streit, "Visual exploration of relationships and structure in low-dimensional embeddings," *IEEE Transactions on Visualization and Computer Graphics*, vol. 29, no. 7, pp. 3312–3326, 2022.
- [16] T. Fujiwara, O.-H. Kwon, and K.-L. Ma, "Supporting analysis of dimensionality reduction results with contrastive learning," *IEEE transactions on visualization and computer graphics*, vol. 26, no. 1, pp. 45–55, 2019.
- [17] B. C. Kwon, H. Kim, E. Wall, J. Choo, H. Park, and A. Endert, "Axisketcher: Interactive nonlinear axis mapping of visualizations through user drawings," *IEEE transactions on visualization and computer graphics*, vol. 23, no. 1, pp. 221–230, 2016.
- [18] R. Faust, D. Glickenstein, and C. Scheidegger, "Dimreader: Axis lines that explain non-linear projections," *IEEE transactions on visualization and computer graphics*, vol. 25, no. 1, pp. 481–490, 2018.
- [19] T. Fujiwara, X. Wei, J. Zhao, and K.-L. Ma, "Interactive dimensionality reduction for comparative analysis," *IEEE Transactions on Visualization and Computer Graphics*, vol. 28, no. 1, pp. 758–768, 2021.
- [20] H. Han, R. Faust, B. F. Keith Norambuena, J. Lin, S. Li, and C. North, "Explainable interactive projections of images," *Machine Vision and Applications*, vol. 34, no. 6, p. 100, 2023.
- [21] J. Xia, W. Lin, G. Jiang, Y. Wang, W. Chen, and T. Schreck, "Visual clustering factors in scatterplots," *IEEE Computer Graphics and Applications*, vol. 41, no. 5, pp. 79–89, 2021.
- [22] G. Appleby, M. Espadoto, R. Chen, S. Goree, A. C. Telea, E. W. Anderson, and R. Chang, "Hypernp: Interactive visual exploration of multidimensional projection hyperparameters," in *Computer Graphics Forum*, vol. 41, no. 3. Wiley Online Library, 2022, pp. 169–181.
- [23] S. Liu, D. Maljovec, B. Wang, P.-T. Bremer, and V. Pascucci, "Visualizing high-dimensional data: Advances in the past decade," *IEEE transactions on visualization and computer graphics*, vol. 23, no. 3, pp. 1249–1268, 2016.
- [24] H. Hotelling, "Analysis of a complex of statistical variables into principal components," *Journal of educational psychology*, vol. 24, no. 6, p. 417, 1933.
- [25] P. Xanthopoulos, P. M. Pardalos, T. B. Trafalis, P. Xanthopoulos, P. M. Pardalos, and T. B. Trafalis, "Linear discriminant analysis," *Robust data mining*, pp. 27–33, 2013.
- [26] D. H. Jeong, C. Ziemkiewicz, B. Fisher, W. Ribarsky, and R. Chang, "ipca: An interactive system for pca-based visual analytics," in *Computer Graphics Forum*, vol. 28, no. 3. Wiley Online Library, 2009, pp. 767–774.
- [27] J. Choo, H. Lee, J. Kihm, and H. Park, "ivisclassifier: An interactive visual analytics system for classification based on supervised dimension reduction," in *2010 IEEE Symposium on Visual Analytics Science and Technology*. IEEE, 2010, pp. 27–34.
- [28] J. B. Tenenbaum, V. d. Silva, and J. C. Langford, "A global geometric framework for nonlinear dimensionality reduction," *science*, vol. 290, no. 5500, pp. 2319–2323, 2000.
- [29] W. S. Torgerson, *Theory and methods of scaling*. Wiley, 1958.
- [30] K. Q. Weinberger, F. Sha, and L. K. Saul, "Learning a kernel matrix for nonlinear dimensionality reduction," in *Proceedings of the twenty-first international conference on Machine learning*, 2004, p. 106.
- [31] M. Espadoto, N. S. T. Hirata, and A. C. Telea, "Deep learning multidimensional projections," *Information Visualization*, vol. 19, no. 3, pp. 247–269, 2020.
- [32] Z. Zang, S. Cheng, L. Lu, H. Xia, L. Li, Y. Sun, Y. Xu, L. Shang, B. Sun, and S. Z. Li, "Evnet: An explainable deep network for dimension reduction," *IEEE Transactions on Visualization and Computer Graphics*, 2022.
- [33] A. Hinterreiter, C. Humer, B. Kainz, and M. Streit, "Paradime: A framework for parametric dimensionality reduction," in *Computer Graphics Forum*, vol. 42, no. 3. Wiley Online Library, 2023, pp. 337–348.
- [34] D. Sacha, L. Zhang, M. Sedlmair, J. A. Lee, J. Peltonen, D. Weiskopf, S. C. North, and D. A. Keim, "Visual interaction with dimensionality reduction: A structured literature analysis," *IEEE transactions on visualization and computer graphics*, vol. 23, no. 1, pp. 241–250, 2016.
- [35] N. Elmqvist, P. Dragicevic, and J.-D. Fekete, "Rolling the dice: Multidimensional visual exploration using scatterplot matrix navigation," *IEEE transactions on Visualization and Computer Graphics*, vol. 14, no. 6, pp. 1539–1148, 2008.
- [36] M. Cavallo and Ç. Demiralp, "A visual interaction framework for dimensionality reduction based data exploration," in *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, 2018, pp. 1–13.
- [37] J. Wenskovitch and C. North, "Observation-level interaction with clustering and dimension reduction algorithms," in *Proceedings of the 2nd Workshop on Human-In-the-Loop Data Analytics*, 2017, pp. 1–6.
- [38] J. Wenskovitch, I. Crandell, N. Ramakrishnan, L. House, and C. North, "Towards a systematic combination of dimension reduction and clustering in visual analytics," *IEEE transactions on visualization and computer graphics*, vol. 24, no. 1, pp. 131–141, 2017.
- [39] E. P. dos Santos Amorim, E. V. Brazil, J. Daniels, P. Joia, L. G. Nonato, and M. C. Sousa, "ilamp: Exploring high-dimensional spacing through backward multidimensional projection," in *2012 IEEE Conference on Visual Analytics Science and Technology (VAST)*. IEEE, 2012, pp. 53–62.
- [40] J. Z. Self, M. Dowling, J. Wenskovitch, I. Crandell, M. Wang, L. House, S. Leman, and C. North, "Observation-level and parametric interaction for high-dimensional data analysis," *ACM Transactions on Interactive Intelligent Systems (TiiS)*, vol. 8, no. 2, pp. 1–36, 2018.
- [41] M. Dowling, J. Wenskovitch, J. Fry, L. House, and C. North, "Sirius: Dual, symmetric, interactive dimension reductions," *IEEE transactions on visualization and computer graphics*, vol. 25, no. 1, pp. 172–182, 2018.
- [42] S. Johansson and J. Johansson, "Interactive dimensionality reduction through user-defined combinations of quality metrics," *IEEE transactions on visualization and computer graphics*, vol. 15, no. 6, pp. 993–1000, 2009.

- [43] D. J. Lehmann and H. Theisel, "Optimal sets of projections of high-dimensional data," *IEEE transactions on visualization and computer graphics*, vol. 22, no. 1, pp. 609–618, 2015.
- [44] T. Fujiwara, Y.-H. Kuo, A. Ynnerman, and K.-L. Ma, "Feature learning for nonlinear dimensionality reduction toward maximal extraction of hidden patterns," in *2023 IEEE 16th Pacific Visualization Symposium (PacificVis)*. IEEE, 2023, pp. 122–131.
- [45] A. Bardos, I. Mollas, N. Bassiliades, and G. Tsoumakas, "Local explanation of dimensionality reduction," in *Proceedings of the 12th Hellenic conference on artificial intelligence*, 2022, pp. 1–9.
- [46] N. Mylonas, I. Mollas, N. Bassiliades, and G. Tsoumakas, "Exploring local interpretability in dimensionality reduction: Analysis and use cases," *Expert Systems with Applications*, vol. 252, p. 124074, 2024.
- [47] X. Zeng, H. Zhou, Z. Li, C. Zhang, J. Lin, J. Xia, Y. Yang, and X. Kui, "ihelp: interactive hierarchical linear projections for interpreting non-linear projections," *Journal of Visualization*, vol. 26, no. 3, pp. 631–648, 2023.
- [48] A. Ghosh, M. Nashaat, J. Miller, and S. Quader, "Interpretation of structural preservation in low-dimensional embeddings," *IEEE Transactions on Knowledge and Data Engineering*, vol. 34, no. 5, pp. 2227–2240, 2020.
- [49] O. Ovcharenko, R. Sevastjanova, and V. Boeva, "Feature clock: High-dimensional effects in two-dimensional plots," *arXiv preprint arXiv:2408.01294*, 2024.
- [50] J. Li and C.-q. Zhou, "Incorporation of human knowledge into data embeddings to improve pattern significance and interpretability," *IEEE Transactions on Visualization and Computer Graphics*, vol. 29, no. 1, pp. 723–733, 2022.
- [51] B. Montambault, G. Appleby, J. Rogers, C. D. Brumar, M. Li, and R. Chang, "Dimbridge: Interactive explanation of visual patterns in dimensionality reductions with predicate logic," *IEEE Transactions on Visualization and Computer Graphics*, 2024.
- [52] Y. Bian, C. North, E. Krokos, and S. Joseph, "Semantic explanation of interactive dimensionality reduction," in *2021 IEEE Visualization Conference (VIS)*. IEEE, 2021, pp. 26–30.
- [53] J. Stahnke, M. Dörk, B. Müller, and A. Thom, "Probing projections: Interaction techniques for interpreting arrangements and errors of dimensionality reductions," *IEEE transactions on visualization and computer graphics*, vol. 22, no. 1, pp. 629–638, 2015.
- [54] A. Ghosh, M. Nashaat, J. Miller, and S. Quader, "Visexpres: A visual interactive toolkit for user-driven evaluations of embeddings," *IEEE Transactions on Visualization and Computer Graphics*, vol. 28, no. 7, pp. 2791–2807, 2020.
- [55] A. Chatzimpampas, R. M. Martins, and A. Kerren, "t-visne: Interactive assessment and interpretation of t-sne projections," *IEEE transactions on visualization and computer graphics*, vol. 26, no. 8, pp. 2696–2714, 2020.
- [56] S. Lundberg, "A unified approach to interpreting model predictions," *arXiv preprint arXiv:1705.07874*, 2017.
- [57] M. T. Ribeiro, S. Singh, and C. Guestrin, "why should i trust you?" explaining the predictions of any classifier," in *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*, 2016, pp. 1135–1144.
- [58] M. Wattenberg, F. Viégas, and I. Johnson, "How to use t-sne effectively," *Distill*, 2016. [Online]. Available: <http://distill.pub/2016/misread-tsne>
- [59] A. P. Andy Coenen. Understanding umap. [Online]. Available: <https://pair-code.github.io/understanding-umap/>
- [60] K. Wongsuphasawat, D. Moritz, A. Anand, J. Mackinlay, B. Howe, and J. Heer, "Voyager: Exploratory analysis via faceted browsing of visualization recommendations," *IEEE transactions on visualization and computer graphics*, vol. 22, no. 1, pp. 649–658, 2015.
- [61] S. Lloyd, "Least squares quantization in pcm," *IEEE transactions on information theory*, vol. 28, no. 2, pp. 129–137, 1982.
- [62] T. Caliński and J. Harabasz, "A dendrite method for cluster analysis," *Communications in Statistics-theory and Methods*, vol. 3, no. 1, pp. 1–27, 1974.
- [63] M. Sips, B. Neubert, J. P. Lewis, and P. Hanrahan, "Selecting good views of high-dimensional data using class consistency," in *Computer Graphics Forum*, vol. 28, no. 3. Wiley Online Library, 2009, pp. 831–838.
- [64] J. Venna and S. Kaski, "Local multidimensional scaling," *Neural Networks*, vol. 19, no. 6–7, pp. 889–899, 2006.
- [65] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.
- [66] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A next-generation hyperparameter optimization framework," in *Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining*, 2019, pp. 2623–2631.
- [67] J. Bergstra, R. Bardenet, Y. Bengio, and B. Kégl, "Algorithms for hyper-parameter optimization," *Advances in neural information processing systems*, vol. 24, 2011.
- [68] Z. Zang, S. Cheng, H. Xia, L. Li, Y. Sun, Y. Xu, L. Shang, B. Sun, and S. Z. Li, "Dmt-ev: An explainable deep network for dimension reduction," *IEEE Transactions on Visualization and Computer Graphics*, vol. 30, no. 3, pp. 1710–1727, 2022.
- [69] L. Van Der Maaten, E. O. Postma, H. J. Van Den Herik *et al.*, "Dimensionality reduction: A comparative review," *Journal of Machine Learning Research*, vol. 10, no. 66–71, p. 13, 2009.
- [70] J. A. Lee and M. Verleysen, "Quality assessment of dimensionality reduction: Rank-based criteria," *Neurocomputing*, vol. 72, no. 7–9, pp. 1431–1443, 2009.
- [71] F. V. Paulovich, L. G. Nonato, R. Minghim, and H. Levkowitz, "Least square projection: A fast high-precision multidimensional projection technique and its application to document mapping," *IEEE Transactions on Visualization and Computer Graphics*, vol. 14, no. 3, pp. 564–575, 2008.
- [72] H. Jeon, A. Cho, J. Jang, S. Lee, J. Hyun, H.-K. Ko, J. Jo, and J. Seo, "Zadu: A python library for evaluating the reliability of dimensionality reduction embeddings," in *2023 IEEE Visualization and Visual Analytics (VIS)*. IEEE, 2023, pp. 196–200.
- [73] N. C. Platform. Cloud data box / detailed description of provided data. [Online]. Available: <https://guide.ncloud-docs.com/docs/en/clouddatabox-data>
- [74] M. Sedlmair, A. Tatu, T. Munzner, and M. Tory, "A taxonomy of visual cluster separation factors," in *Computer graphics forum*, vol. 31, no. 3pt4. Wiley Online Library, 2012, pp. 1335–1344.
- [75] P. Saraiya, C. North, and K. Duca, "An insight-based methodology for evaluating bioinformatics visualizations," *IEEE transactions on visualization and computer graphics*, vol. 11, no. 4, pp. 443–456, 2005.
- [76] D. A. Norman, "The psychology of everyday things," 1988.
- [77] D. Pelleg, A. Moore *et al.*, "X-means: Extending k-means with efficient estimation of the number of clusters," in *ICML'00*. Citeseer, 2000, pp. 727–734.
- [78] B. C. Kwon, B. Eysenbach, J. Verma, K. Ng, C. De Filippi, W. F. Stewart, and A. Perer, "Clustervision: Visual supervision of unsupervised clustering," *IEEE transactions on visualization and computer graphics*, vol. 24, no. 1, pp. 142–151, 2017.



Aeri Cho is a Ph.D. Student at the Department of Computer Science and Engineering, Seoul National University, Korea. Her research interests are in Interactive Data Analysis and Visual Analytics. Before starting her Ph.D. program, she received a B.S. degree in Computer Science and Engineering from Seoul National University.



Hyeon Jeon is a Ph.D. Student at the Department of Computer Science and Engineering, Seoul National University, Korea. His research interests span the field of Visual Analytics and Machine Learning. Before starting his Ph.D. program, he received a B.S. degree in Computer Science and Engineering from POSTECH.



Kiroong Choe is a Ph.D. Student at the Department of Computer Science and Engineering, Seoul National University, Korea. His research interests include Human-AI Collaboration, Educational System Design, and Information Visualization. Before starting his Ph.D. program, he received a B.S. and M.S. degrees from Seoul National University.



Seokhyeon Park is a Ph.D. Student at the Department of Computer Science and Engineering, Seoul National University, Korea. His research interests include Human-AI Interaction and Interface Design. Before starting his Ph.D. program, he received a B.S. degree in Computer Science and Engineering and a B.A. degree in Information Science and Culture from Seoul National University.



Jinwook Seo is a professor in the Department of Computer Science and Engineering, Seoul National University, where he is also the Director of the Human-Computer Interaction Laboratory. His research interests include Human-Computer Interaction, Information Visualization, and Biomedical Informatics. He received his PhD in Computer Science from the University of Maryland at College Park in 2005.